

SIYUAN LIU

Computational Solid Mechanics | Nonlinear Materials | Multiphysics Simulation

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RESEARCH PROFILE

Ph.D. candidate in Chemical Engineering specializing in computational mechanics and multiphysics modeling of high-temperature electrochemical cell systems. Research integrates elastoplasticity, cohesive-zone debonding, phase-field fracture, Peridynamics(NOSB-PD), electrochemical transport, and nonlinear FE solution strategies. Experience developing and verifying custom models in MOOSE and Python, with broader simulation experience in COMSOL Multiphysics and ANSYS Fluent. Current work targets thermo-chemo-mechanical degradation, residual stress, redox-induced deformation, interface failure, and brittle fracture in solid oxide and protonic ceramic cells.

EDUCATION

West Virginia University — Ph.D. Candidate, Chemical Engineering 2023–Present

GPA: 3.66/4.00 | Research focus: coupled mechanics, fracture, and electrochemical degradation in solid oxide/protonic ceramic cells

China University of Mining & Technology — M.S., Safety Science and Engineering 2019–2022

Thesis: Dynamic evolution mechanism of high-pressure water and gas alternating sequestration in low-permeability coal seams

Taiyuan University of Technology — B.S., Safety Engineering 2015–2019

RESEARCH PROJECTS

Redox-Induced stress and Fracture on BZCY Electrolyte WVU | 2026

MOOSE finite-element framework | FFT | **phase-field fracture** | Peridynamics

- Developed a microstructure-informed finite-element framework to investigate chemically induced deformation, interfacial degradation, and fracture in ceramic–metal energy materials.
- Formulated and coupled nonlinear constitutive, interfacial, and fracture models, building experience in finite-strain mechanics, inelastic material response, and failure analysis.
- Designed verification and numerical-credibility studies address constitutive response, fracture irreversibility, nonlinear convergence, discretization sensitivity, and uncertain model parameters.
- Connected local stress redistribution and damage evolution with microstructural features and degradation mechanisms, establishing a framework for durability, cyclic loading, and lifetime-oriented studies.

Physics-Based Modeling of Electronic Leakage and Carrier Selectivity in Bilayer Protonic WVU | 2025

Ceramic Electrolysis Cells

Electrochemical modeling | nonlinear transport | numerical solvers | parameter screening

- Developed voltage- and current-driven electrochemical models for proton transport, oxygen-vacancy conduction, and parasitic hole leakage in bilayer protonic ceramic electrolytes.
- Introduced a Poole–Frenkel field-assisted hole-transport formulation resolving nonlinear conductivity, local electric-field redistribution, and electronic-current continuity across electrolyte layers.
- Established a current-controlled parallel-circuit framework predicting carrier-current splitting, electrolyte polarization, terminal voltage, hydrogen production, and Faradaic efficiency.
- Performed material- and geometry-parameter sweeps to identify trade-offs between protonic resistance and electronic blocking; implemented conservation checks and automated numerical result generation in Python.

RESEARCH & ENGINEERING EXPERIENCE

- Research Assistant, West Virginia University

2023–Present

 - Model coupled mechanical, chemical, thermal, and electrochemical phenomena governing structural stability and degradation of solid oxide and protonic ceramic cells.
 - Develop custom numerical workflows for nonlinear finite-element analysis, fracture and interface failure, electrochemical transport, parameter studies, verification, and post-processing.
- Safety Evaluation Engineer, China Institute for Radiation Protection

2022–2023

 - Performed safety evaluations of industrial facilities, identifying failure modes and operational risks and supporting engineering decisions through systematic hazard assessment.
- Research Assistant, National Natural Science Foundation of China (Grant 52174211)

2021–2022

 - Built and validated multiphase CFD models for pore-scale/continuum transport relevant to coalbed methane and shale-gas extraction, strengthening experience in transport physics and numerical verification.
- Principal Investigator, Jiangsu Postgraduate Research & Practice Innovation Program

2021–2022

 - Led a research project on equipment and process design for high-pressure water–gas alternating sequestration technology.

COMPUTATIONAL & RESEARCH SKILLS

Mechanics	Finite-strain mechanics; nonlinear constitutive modeling; J2 plasticity; fracture mechanics; phase-field fracture; cohesive-zone modeling; residual stress; eigenstrain; interface debonding
Finite Elements	MOOSE; COMSOL Multiphysics; nonlinear Newton solution; variational inequalities; mesh/convergence studies; model verification
Scientific Computing	Python; C++/MOOSE input and model development; numerical solvers; parameter sweeps; automated post-processing; conservation and verification checks
Multiphysics	Electrochemical transport; porous-media/multiphase flow; heat and mass transfer; CFD; thermo-chemo-mechanical coupling; solid oxide/protonic ceramic energy systems

SELECTED PUBLICATIONS & INTELLECTUAL PROPERTY

- Liu, S., et al. **Multiphysics Modeling of Mechanical Failure in Solid Oxide Cells: Methodology and Progress**. International Journal of Hydrogen Energy, under review.
- Liu, S., et al. **Coupled Proton–Electron Transport and Electronic Leakage in Proton-Conducting Electrolysis Cells: Modeling and Multilayer Electrolyte Design**. Manuscript in preparation.
- Zhu C., Liu S., Chen X., et al. **High-pressure water and gas alternating sequestration technology for low-permeability coal seams with high adsorption capacity**. Journal of Natural Gas Science and Engineering (2021).
- Li K., Zhu C., Liu S., et al. **Effects of injection pressure and duration on alternate high-pressure water–gas sequestration of coalbed methane**. Geofluids (2022).
- Liu S., Zhu C., et al. **Mechanism of adsorption capacity enhancement of coal due to interaction with a high-pressure scCO₂–water system**. International Journal of Greenhouse Gas Control (2023).
- Patent: A recyclable high-pressure drilling sealing device with high resistance and its use method. CN111395991B.
- Patent: A water-cycle internal slag discharge drill pipe. CN213330951U.

HONORS

First Prize Scholarship, China University of Mining & Technology (Top 20%, 2021) | National Encouragement Scholarship, Taiyuan University of Technology (Top 5%, 2018)